

Math 61: Introduction to Discrete Structures
Discussion 9: Graph Theory

1 Dijkstra

From my old prof. <https://homes.cs.washington.edu/~anuprao/pubs/CSE421Wi2020/DijkstraBF.pdf>.

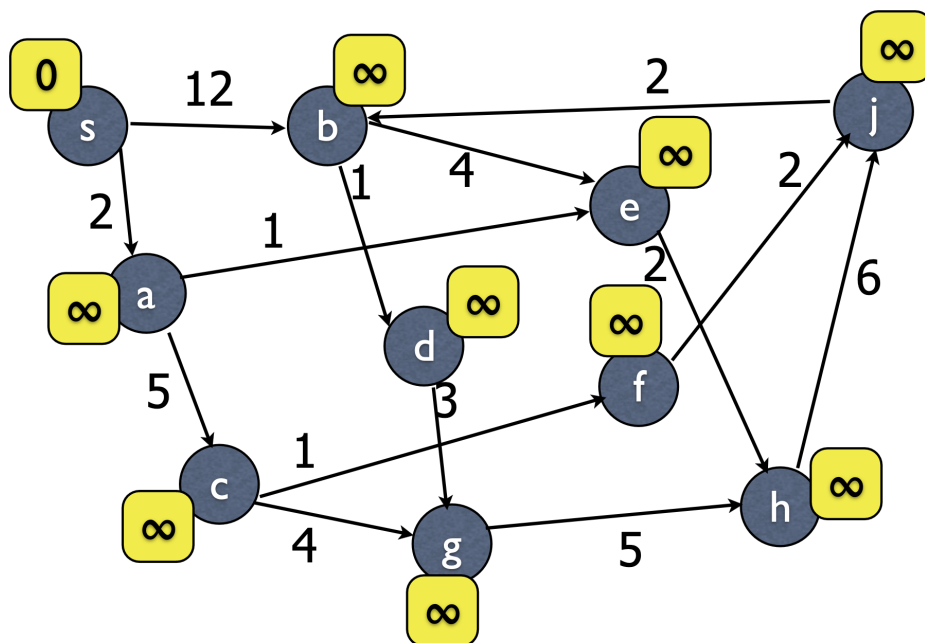
Dijkstra(s)

Set all vertices v undiscovered, $d(v) = \infty$

Set $d(s) = 0$, mark s discovered.

while there is edge from discovered vertex to undiscovered vertex,

 let (u,v) be such edge minimizing $d(u) + l_{u,v}$
 set $d(v) = d(u) + l_{u,v}$, mark v discovered



Question1: in the above graph, run the entire Dijkstra algorithm (the lengths have already been initialized to 0 and ∞). (See the first couple pages of <https://homes.cs.washington.edu/~anuprao/pubs/CSE421Wi2020/DijkstraBF.pdf> for the answers.)

Question2: try to write a ≤ 30 word summary/outline of the key points of the proof of correctness for Dijkstra's algorithm. (One possible answer: <https://homes.cs.washington.edu/~anuprao/pubs/CSE421Wi2020/DijkstraBF.pdf#page=23>)

Question3 (8.4.10): True or false? When a connected, weighted graph and vertices a and z are input to the following algorithm, it returns the length of a shortest path from a to z . If the algorithm is correct, prove it; otherwise, give an example of a connected, weighted graph and vertices a and z for which it fails. (Hint: can consider graphs with 3 vertices)

Algorithm 8.4.6

```
algor( $w, a, z$ ) {  
    length = 0  
     $v = a$   
     $T = \text{set of all vertices}$   
    while ( $v \neq z$ ) {  
         $T = T - \{v\}$   
        choose  $x \in T$  with minimum  $w(v, x)$   
        length = length +  $w(v, x)$   
         $v = x$   
    }  
    return length  
}
```

Question4 (8.4.11): True or false? Dijkstra's algorithm finds the length of a shortest path in a connected, weighted graph even if some weights are negative. If true, prove it; otherwise, provide a counter-example. (Hint: can consider graphs with 3 vertices)

2 Adjacency Matrix

Example 8.5.1

Go Online

For more on the adjacency matrix, see goo.gl/jfjmoI

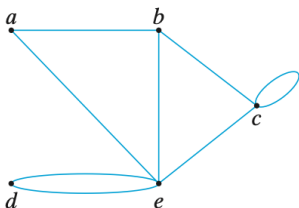


Figure 8.5.1 The graph for Example 8.5.1.

Adjacency Matrix Consider the graph of Figure 8.5.1. To obtain the adjacency matrix of this graph, we first select an ordering of the vertices, say a, b, c, d, e . Next, we label the rows and columns of a matrix with the ordered vertices. The entry in this matrix in row i , column j , $i \neq j$, is the number of edges incident on i and j . If $i = j$, the entry is twice the number of loops incident on i . The adjacency matrix for this graph is

$$\begin{array}{c} a \quad b \quad c \quad d \quad e \\ \begin{array}{l} a \\ b \\ c \\ d \\ e \end{array} \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 2 \\ 1 & 1 & 1 & 2 & 0 \end{pmatrix} \end{array}.$$

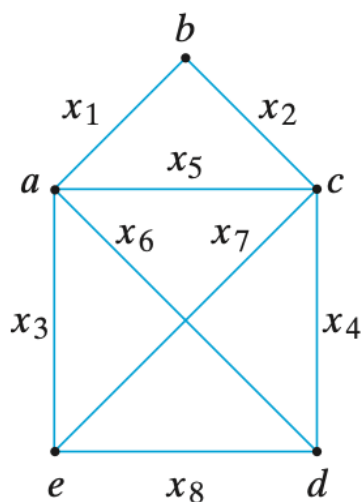
Notice that we can obtain the degree of a vertex v in a graph G by summing row v or column v in G 's adjacency matrix.

The adjacency matrix is not a very efficient way to represent a graph. Since the matrix is symmetric about the main diagonal (the elements on a line from the upper-left corner to the lower-right corner), the information, except that on the main diagonal, appears twice.

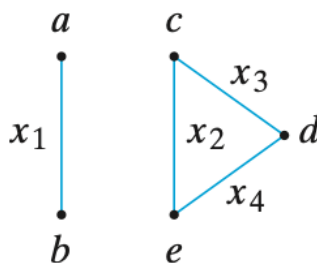
See next page for some problems.

In Exercises 1–6, write the adjacency matrix of each graph.

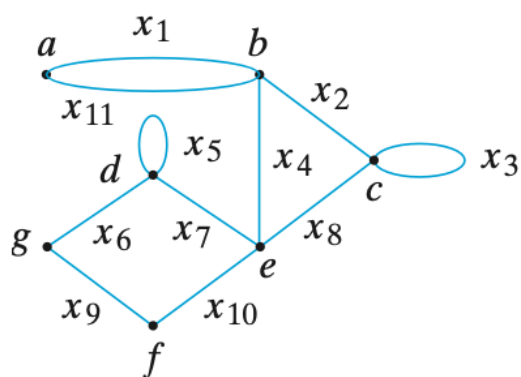
1.



2.



3.



24.
$$\begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

25.
$$\begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

26. What must a graph look like if some row of its incidence matrix consists only of 0's?